

CHEMISTRY (CODE - 043)
SAMPLE QUESTION PAPER*
CLASS XII (2025-26)

Time: 3 hours

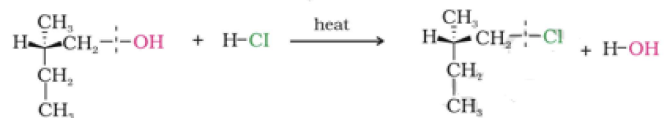
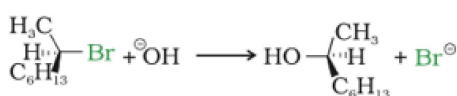
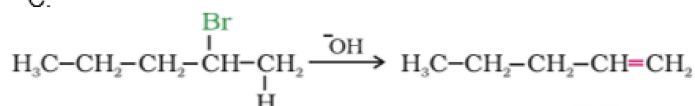
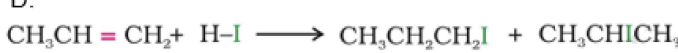
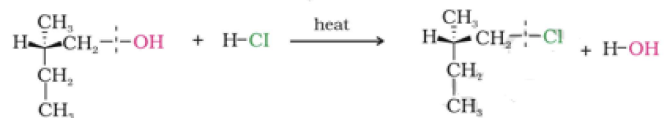
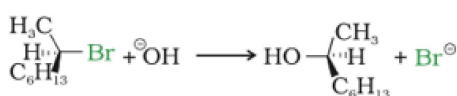
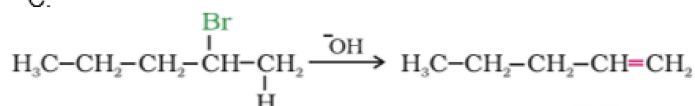
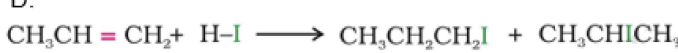
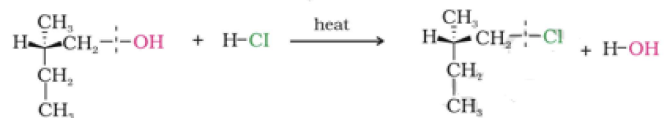
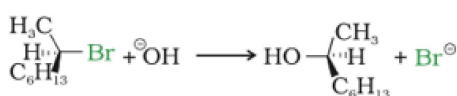
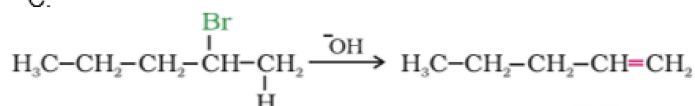
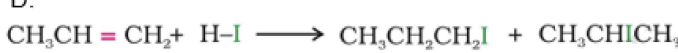
Max. Marks: 70

GENERAL INSTRUCTIONS:

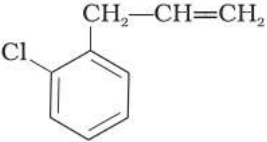
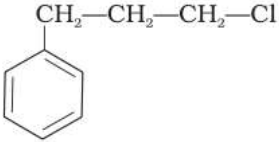
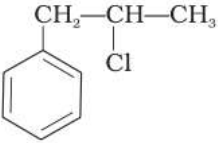
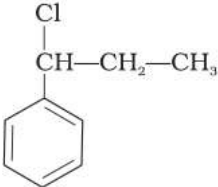
Read the following instructions carefully.

1. There are **33** questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. SECTION B consists of 5 short answer questions carrying 2 marks each.
4. SECTION C consists of 7 short answer questions carrying 3 marks each.
5. SECTION D consists of 2 case-based questions carrying 4 marks each.
6. SECTION E consists of 3 long answer questions carrying 5 marks each.
7. All questions are compulsory.
8. Use of log tables and calculators is not allowed.

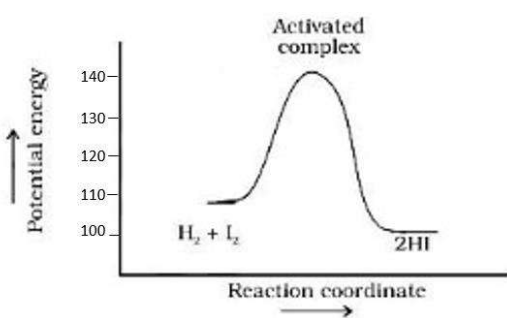
Section-A		
Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.		
1	Which of the following reaction will lead to formation of ethyl methyl ketone: A. heating $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$ with acidified $\text{Na}_2\text{Cr}_2\text{O}_7$ B. passing $\text{CH}_3\text{C}(\text{OH})\text{CH}_3$ over heated copper C. ozonolysis of $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_3$ D. acetylene on reaction with $\text{HgSO}_4/\text{H}_2\text{SO}_4$	1
2	Consider the reaction and identify B and C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl} \xrightarrow{\text{NaOH + Ethanol}} \text{A}$ $\text{A} \xrightarrow{\text{H}_2\text{O}, \text{H}^+} \text{B}$ and $\text{A} \xrightarrow{\text{(i) B}_2\text{H}_6, \text{(ii) H}_2\text{O}_2, \text{OH}^-} \text{C}$ A. B=C= Butanol B. B= Butanol, C=Butene C. B= Butan-2-ol, C= Butanol D. B= Butene, C=Butan-2-ol	1
3	The counter ion in the coordination compound $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)] \text{Cl}_2$ is A. Ammine B. Cobalt	1

	C. Chloride D. Nitro											
4	The organic compounds A, B and C are amines having equivalent molecular weight. A and B on reaction with benzene sulphonyl chloride give white precipitate, however white precipitate obtained from compound B remains insoluble in NaOH. The variation in the boiling point of A, B and C can be seen as : A. $A > B > C$ B. $B > A > C$ C. $A = B > C$ D. $C > B > A$	1										
5	70 gm solute is dissolved in 700 gm solvent to prepare a solution having density 1.5 g/ml. The ratio of its molality and molarity will be: A. 0.77 B. 1.4 C. 0.73 D. 1.3	1										
6	Match the column I and column II: <table><thead><tr><th>Column I</th><th>Column II</th></tr></thead><tbody><tr><td>A. </td><td>(i) Addition reaction</td></tr><tr><td>B. </td><td>(ii) Elimination reaction</td></tr><tr><td>C. </td><td>(iii) S_N^2 reaction</td></tr><tr><td>D. </td><td>(iv) S_N^1 reaction</td></tr></tbody></table> A. A-(i), B-(ii), C-(iii), D-(iv) B. A-(iv), B-(ii), C-(iii), D-(i) C. A-(i), B-(iii), C-(ii), D-(iv) D. A-(iv), B-(iii), C-(ii), D-(i)	Column I	Column II	A. 	(i) Addition reaction	B. 	(ii) Elimination reaction	C. 	(iii) S_N^2 reaction	D. 	(iv) S_N^1 reaction	1
Column I	Column II											
A. 	(i) Addition reaction											
B. 	(ii) Elimination reaction											
C. 	(iii) S_N^2 reaction											
D. 	(iv) S_N^1 reaction											

7	In o-cresol, the –OH group is attached to the carbon that is: A. sp^3 hybrid B. sp^2 hybrid C. sp hybrid D. dsp^2 hybrid	1
8	Which of the following is laevorotatory in nature : A. alpha D – glucose B. beta -D- glucose C. beta-D- fructose D. sucrose	1
9	The name inner transition metals is often used to refer to the A. lanthanoids B. actinoids C. both lanthanoids and actinoids D. d block elements	1
10	Λ_m^o CH₃COOH can be calculated if the values of the following are given: 1. Λ_m^o HCl, Λ_m^o KCl and Λ_m^o CH₃COOK 2. Λ_m^o NaCl, Λ_m^o KCl and Λ_m^o CH₃COONa 3. Λ_m^o H₂SO₄, Λ_m^o Na₂SO₄ and Λ_m^o CH₃COONa A. Only 1 B. Either 1 or 2 C. Either 1 or 3 D. Either 2 or 3	1
11	Which of the following will give a yellow or orange ppt. with 2,4 DNP? (i) Propanal (ii) Propanone (iii) Propanoic acid A. (i) and (ii) B. (ii) and (iii) C. (i) and (iii) D. (i), (ii) and (iii)	1

12	Identify the secondary benzylic halide and primary alkyl halide from the following: (i)  (ii)  (iii)  (iv)  A. (i) and (iii) B. (iv) and (ii) C. (iii) and (iv) D. (i) and (ii)	1
13	Assertion (A): Secondary aliphatic amines react with nitrous acid to form aliphatic diazonium salts which being unstable, liberate nitrogen gas quantitatively. Reason: Nitrogen gas evolved on reaction with nitrous acid can be used for the estimation of proteins and amino acids. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true	1
14	Assertion (A): Care should always be taken to ensure that solutions that flow in the blood stream are of the same osmotic pressure as that of the blood. Reason(R): Sodium ion and potassium ions are responsible for maintaining proper osmotic pressure balance inside and outside of the cells of organism. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true	1

15	Assertion (A): Starch forms colloidal solution with water. Reason (R): Starch contains 80-85% of amylopectin which is insoluble in water. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true	1
16	Assertion (A): Secondary cells are used in invertors. Reason (R): A primary cell can be recharged by passing current through it in the opposite direction after it has been used. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true	1
Section-B Question No. 17 to 21 are very short answer questions carrying 2 marks each.		
17	Attempt either option A or B A. Answer the following: - When 50 mL of phenol and 50 mL of aniline are mixed, predict whether the volume of the solution is equal to, greater than or less than 100 mL. Give reason to support your answer. - Ritesh suggested adding salt to the box containing ice. He said this would keep the cold drink bottles cold for a longer time. How will Ritesh justify his suggestion? OR B. Answer the following: BaCl_2 on reaction with Na_2SO_4 in aqueous solution gives white precipitate. If the two solutions are separated by a semi-permeable membrane, will there be appearance of a white precipitate due to osmosis? - Why does water stops boiling when sugar is added to boiling water.	2x1 2x1

18	Consider the graph for the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$  I. Calculate enthalpy of reaction and activation energy for the backward reaction. II. How will the catalyst affect the rate of this reaction? Explain. (for visually challenged learners) I. Define activation energy. What will happen to activation energy if we increase the temperature? II. How will the catalyst affect the rate of an endothermic reaction?	2x1 2x1
19	Carry out following conversions : I. Nitrobenzene to 4- bromobenzenamine II. Chlorophenylmethane to 2-phenyl-ethanamine	2x1
20	Write the formula of the following coordination complex: I. diaquasilver(I) dichloridoargentate(I) II. dihydroxidobis(triphenylphosphine)nickel(II)	2x1
21	The mechanism of formation of alcohols from alkenes is given below. Rectify the errors in the mechanism and rewrite the corrected steps STEP 1 $>\text{C}=\text{C}< + \text{H}-\overset{\text{H}}{\underset{\cdot\cdot}{\text{O}}}-\text{H} \rightleftharpoons \begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}^+- \\ \end{array} + \text{H}_2\ddot{\text{O}}$ STEP 2 $\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}^+- \\ \end{array} + \text{H}_2\ddot{\text{O}} \rightleftharpoons \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}-\text{O}^+-\text{H} \end{array}$ STEP 3 $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}-\text{O}^+-\text{H} \end{array} + \text{H}_2\ddot{\text{O}} \rightarrow \begin{array}{c} \text{H} \quad \text{OH} \\ \quad \\ -\text{C}-\text{C}- \\ \end{array} + \text{H}_3\text{O}^+$	2

Section-C

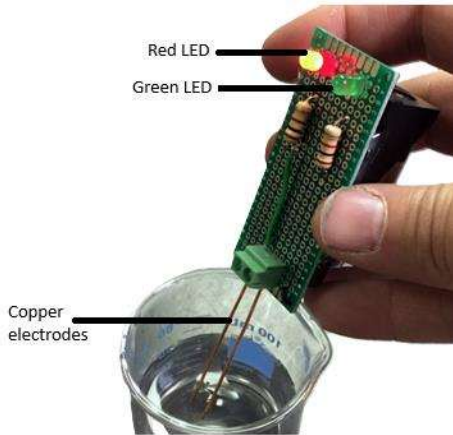
Question No. 22 to 28 are short answer questions, carrying 3 marks each.

22	What will be the vapour pressure of a 1 molal aqueous solution of MgCl_2 , assuming dissociation of MgCl_2 to be 70 mole percent? (Vapor pressure of pure water at 25 °C is 23.8 mmHg)	3
23	Write the Nernst equation for the following: I. $\text{Ni (s)} + \text{Cu}^{2+} \text{ (aq)} \rightarrow \text{Ni}^{2+} \text{ (aq)} + \text{Cu (s)}$ II. $\text{Al (s)} + \text{FeSO}_4 \text{ (aq)} \rightarrow \text{Al}_2(\text{SO}_4)_3 \text{ (aq)} + \text{Fe (s)}$ III. $\text{Mg (s)}/\text{Mg}^{2+} \text{ (aq)}/\text{Ag}^+ \text{ (aq)}/\text{Ag(s)}$	3x1
24	Explain the following: I. Toluene on treatment with Cl_2 in sunlight gives benzyl chloride whereas when treated with Cl_2 in dark gives o-chlorobenzene and p-chlorobenzene. II. Finkelstein reaction is carried out in the presence of dry acetone. III. neo pentylchloride has lower boiling point than isopentylchloride.	3x1
25	Which of the following elements will: I. exhibit similar magnetic behaviour and why? Magnesium (Atomic No. 12), Chromium (Atomic No. 24), Iron (Atomic No. 26) and Molybdenum (Atomic No. 42). II. form white salts and why? Zinc (Atomic No. 30), Scandium (Atomic No. 21), Nickel (Atomic No. 28) and Vanadium (Atomic No. 23)	2x1.5
26	Arrange the products obtained in the following cases in the increasing order of their pKa values: A. Oxidation of ethanol in presence of acidified potassium dichromate B. Reaction of propanoic acid with Br_2 in the presence of red Phosphorus C. Reaction of isopropyl magnesium bromide with carbon dioxide, followed by hydrolysis. D. Reaction of propanoic acid with Cl_2 in the presence of red Phosphorous.	3

27	Carry out the following conversions. (Attempt any 3) I. Butan-2-one to 3-Methylpentan-3-ol II. Anisole to 4-Methoxytoluene III. Phenol to Benzene IV. Chloroethane to Ethoxy ethane	3x1						
28	Answer the following questions: I. Are the enthalpies of atomisation of Zinc and Copper matched correctly? Justify your answer. <table border="1"><tr><td>Element</td><td>Enthalpy of atomisation/ kJmol⁻¹</td></tr><tr><td>Zinc</td><td>339</td></tr><tr><td>Copper</td><td>130</td></tr></table> II. Out of sulphuric acid and hydrochloric acid, which acid will you prefer for permanganate titrations and why? III. $5\text{NO}_2^- + 2\text{MnO}_4^- + 6\text{H}^+ \rightarrow$	Element	Enthalpy of atomisation/ kJmol ⁻¹	Zinc	339	Copper	130	3x1
Element	Enthalpy of atomisation/ kJmol ⁻¹							
Zinc	339							
Copper	130							

Section D

Question No. 29 & 30 are case-based/data -based questions carrying 4 marks each.

29	Conductivity of Aqueous Solutions Electrical conductivity is based on the flow of ions. Slightly ionized substances are <i>weak electrolytes</i>. Weak acids and bases would be categorized as weak electrolytes because they do not completely dissociate in solution. Highly ionized substances are <i>strong electrolytes</i>. Strong acids and salts are strong electrolytes because they completely ionize in solution. The ions carry the electric charge through the solution thus creating an electric current. The current, if sufficient enough, will light one or both LEDs on a <i>conductivity meter</i>, shown at right.  The meter has a 9V battery, two parallel copper electrodes and	
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two LED's – one green and one red. The conductivity of a solution can be tested by switching the meter on and dipping the copper electrodes in the solution to be tested.

Substances that do not conduct an electric current are called *non-electrolytes*. Non-electrolytes do not ionize; they do not contain moveable ions. The LEDs of a conductivity meter will not light because there are no ions to carry the electric current.

The table given below is a guide to the possible conductivity measurements

Scale	Red LED	Green LED	Conductivity
0	off	off	low or none
1	dim	off	low
2	medium	off	medium
3	bright	dim	high
4	very bright	medium	very high

source: <https://chem.libretexts.org>

Based on the information provided above, answer the following questions:

1+1+2

- I. Is it possible to identify whether the given solution is 1 M NaOH or 1 M HCl using the conductivity meter? Justify your answer.
- II. What is the possible pH value of solution if the glow of green LED is medium and the red LED glows very brightly?

(i) 1 (ii) 13 (iii) 5 (iv) 8

- (a) (i) and (ii)
- (b) (i) and (iii)
- (c) (ii) and (iv)
- (d) (iii) and (iv)

OR

Write down the observations if the conductivity meter is dipped in distilled water.

- III. Refer to the table given below and draw the molar conductivity vs. concentration curve for solution A and B.

Solution	Red LED	Green LED
A	bright	dim
B	dim	off

(For Visually Challenged students)

Conductivity of Aqueous Solutions

Conductivity meter is a device which is used to identify whether the given solution is a strong, weak or non-electrolyte. The meter has a 9V battery, and two parallel copper electrodes and a 5 point scale (0 to 4) to measure conductivity of a solution. The conductivity of a solution can be tested by switching the meter on and dipping the copper electrodes in the solution to be tested.

Electrical conductivity is based on the flow of ions. Highly ionized substances are *strong electrolytes*. Strong acids and salts are strong electrolytes because they completely ionize in solution. The ions carry the electric charge through the solution thus creating an electric current. The current, if sufficient enough, will show a value of 3 or 4 on the conductivity *meter*.

Slightly ionized substances are *weak electrolytes*. Weak acids and bases would be categorized as weak electrolytes because they do not completely dissociate in solution. The values for weak electrolytes are 1 or 2 on the conductivity scale.

Substances that do not conduct an electric current are called *non-electrolytes*. Non-electrolytes do not ionize; they do not contain moveable ions. The conductivity meter shows a value of 0 in such a case as there are no ions to carry the electric current.

The following table is a guide to the possible conductivity values:

Scale	Conductivity
0	low or none
1	low
2	medium
3	high
4	very high

Based on the information provided above, answer the following questions:

1+1+2

- I. Is it possible to identify whether the given solution is 1 M NaOH or 1 M HCl using the conductivity meter? Justify your answer.
- II. What is the possible pH value of solution if the scale shows the value "4"

(i) 1 (ii) 13 (iii) 5 (iv) 8

- (a) (i) and (ii)
- (b) (i) and (iii)
- (c) (ii) and (iv)
- (d) (iii) and (iv)

OR

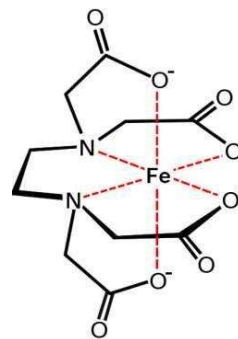
What will be value on the scale if the conductivity meter is dipped in distilled water?

- III. Predict the electrolyte is a strong or weak electrolyte on the basis of the following observation:

Solution	Scale
A	3
B	2

Iron-EDTA complex in food fortification

Food fortification is defined as the practice of adding vitamins and minerals to commonly consumed foods during processing to increase their nutritional value. It is a proven, safe and cost-effective strategy for improving diets and for the prevention and control of micronutrient deficiencies. A food product (such as rice, wheat flour, edible oil) that is fortified through the addition of fortificants is called a “vehicle”.



In African and south Asian countries 40% of the population suffers from anaemia. Average human needs nearly 10mg of iron daily. Iron fortification may be useful in fighting iron deficiencies in humans. Reduced iron and several iron salts have been used in the past as iron fortification, however, not all are suitable for this purpose, in terms of iron absorption. Recent studies have shown that beverages containing sugar fortified with either Ferrous sulphate or Fe(III)- EDTA complex have high rate of absorption of iron.

Ferrous sulphate as well as Fe(III)- EDTA is suitable to enrich sugar, but while iron from ferrous sulphate is precipitated and poorly absorbed when fortified sugar is added to beverages such as tea, Fe(III)- EDTA reacts slowly with tea and iron is not precipitated for at least 24 hr.

Fe(III)-EDTA as iron fortification, has demonstrated so far, more advantages than that observed from other iron salts, including ferrous sulphate. But, EDTA is a chelating agent and its use in food technology to prevent oxidative damage of food has been restricted. Excessive consumption of EDTA can cause abdominal cramps, nausea, low blood pressure and damage to kidneys. According to National Institute of Health, it is unsafe to consume more than 3 g of EDTA per day or continuously for more than 5 to 7 days.

The amount of EDTA necessary for 10 mg of iron fortification, is about 60 mg. This is within the safe limits and is comparable to the usual amount added to the diet.

(source: Layrisse, M., & Martlnez-Torres, C. (1977). Fe (III)-EDTA complex as iron fortification. *The American Journal of Clinical Nutrition*, 30(7), 1166-1174.)

Based on the information provided above, answer the following questions:

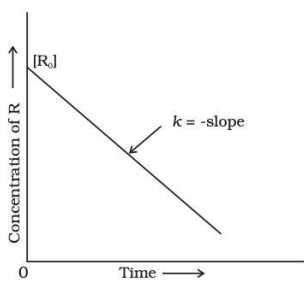
- I. Why is Fe(III)-EDTA complex stable as compared to Ferrous sulphate?

OR

1+1+2

	What happens when hard water is titrated against Na_2EDTA? II. You are a doctor, working in Somalia. Will you recommend iron fortified food to your patients? Support your answer with references from the passage. III. What is the denticity of the ligand in the Fe(III) EDTA complex. Name the atom(s) through which it can bind to the central metal ion. Write the structure of EDTA. (Refer to figure 1) For Visually challenged candidates Iron-EDTA complex in food fortification Food fortification is defined as the practice of adding vitamins and minerals to commonly consumed foods during processing to increase their nutritional value. It is a proven, safe and cost-effective strategy for improving diets and for the prevention and control of micronutrient deficiencies. A food product (such as rice, wheat flour, edible oil) that is fortified through the addition of fortificants is called a “vehicle”. Reduced iron and several iron salts have been used in the past as iron fortification, however, not all are suitable for this purpose, in terms of iron absorption. Recent studies have shown that beverages containing sugar fortified with either Ferrous sulphate or Fe(III)- EDTA complex have high rate of absorption of iron. Ferrous sulphate as well as Fe(III)- EDTA is suitable to enrich sugar, but while iron from ferrous sulphate is precipitated and poorly absorbed when fortified sugar is added to beverages such as tea, Fe(III)- EDTA reacts slowly with tea and iron is not precipitated for at least 24 hr. Fe(III)-EDTA as iron fortification, has demonstrated so far, more advantages than that observed from other iron salts, including ferrous sulphate. But, EDTA is a chelating agent and its use in food technology to prevent oxidative damage of food has been restricted. Excessive consumption of EDTA can cause abdominal cramps, nausea, low blood pressure and damage to kidneys. According to National Institute of Health, it is unsafe to consume more than 3 g of EDTA per day or continuously for more than 5 to 7 days. The amount of EDTA necessary for 10 mg of iron fortification, is about 60 mg. This is within the safe limits and is comparable to the usual amount added to the diet. (source: Layrisse, M., & Martinez-Torres, C. (1977). Fe (III)-EDTA complex as iron fortification. <i>The American Journal of Clinical Nutrition</i>, 30(7), 1166-1174.)	
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	Based on the information provided above, answer the following questions: I. Why is Fe(III)-EDTA complex stable as compared to Ferrous sulphate? OR What happens when hard water is titrated against Na₂EDTA? II. You are a doctor, working in Somalia. Will you recommend iron fortified food to your patients? Support your answer with references from the passage. III. (a) What is the denticity of the ligand in the Fe(III) EDTA complex. Name the atom(s) through which it can bind to the central metal ion. (b) EDTA is an electron acceptor or an electron donor?	1+1+2
Section-E Question No. 31 to 33 are long answer type questions carrying 5 marks each.		
31	Attempt either A or B A. Answer the following questions: I. Write the structure of expected product of Cannizzarro reaction of 2-chlorobenzaldehyde. II. How would the presence of -SO₃H group effect the basic strength of aniline. III. Convert acetic acid to ethanamine. IV. Write the steps to prepare Benzoic acid from Benzoyl chloride. V. Give a chemical test to distinguish between: propanal and propanone OR B. Answer the following questions: I. Write the structure of expected product of Wolf-Kishner reduction of 2 –methylbutanal. II. How would the presence of -SO₃H group effect the acidic strength of benzoic acid III. Prepare acetic acid from ethanamine. IV. Convert Aniline to benzoic acid. V. Give a chemical test to distinguish between: propanal and ethanal.	5x1

32	Attempt either A or B A. Answer the following questions: - Identify and give any one point of difference between the protein present in the hair and protein present in egg albumin. - Both glucose and sucrose have aldehydic group, then why does only glucose reduces Fehling's Reagent, and not sucrose? - Why do amino acids behave as salts? - What chemical change takes place during curdling of milk? - Doctor advised a 50 year old woman enough exposure to sunlight and addition of fish and egg yolk to her diet. What is the possible disease diagnosed by the doctor? OR B. Answer the following questions: - Identify and give any one point of difference between the carbohydrate present in cane sugar and carbohydrate present in milk. - Glucose is an aldohexose and a monosaccharide. Which oxidising agent should be used to bring about oxidation of only the aldehydic group present in glucose? - Amino acid $\text{HOOC-CH}_2\text{CH}(\text{NH}_2)\text{CH}_2\text{COOH}$. Predict whether the pH of this amino acid will be >7, 7 or <7. - Name the two major molecular shapes formed due to the folding of secondary structure of proteins. - Ashish's gums bleed frequently. The doctor's prescription mentioned that Ashish is suffering from scurvy. Help him to identify two food sources to help him recover faster.	5x1
33	Attempt either A or B A. - The rate of a reaction triples when the temperature changes from 298 K to 318 K. Calculate the energy of activation of the reaction assuming that it does not change with temperature. (Given $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$, $\log 3 = 0.4771$) - Identify the order of reaction and write its integrated rate equation mentioning what each term in the equation represents.  OR	3+2

	B. I. Consider the following first order thermal decomposition of SO_2Cl_2 at a constant volume $\text{SO}_2\text{Cl}_2 (\text{g}) \rightarrow \text{SO}_2(\text{g}) + \text{Cl}_2 (\text{g})$ If the total pressure of the gases is found to be 200 torr after 10 seconds and 300 torr upon the complete decomposition of SO_2Cl_2. Calculate the rate constant. (Given $\log 3 = 0.4771$, $\log 2 = 0.3010$) II. For a bimolecular elementary reaction $\text{A} + \text{B} \rightarrow \text{Products}$. Write the expression for the rate of reaction relating temperature and activation energy for the reaction and also mention what each term represents in the equation.	3+2
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